LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Cable system inspection, cable maintenance, fittings, pulleys and wear patterns, Construction and servicing of Bowden cable and Teleflex con
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Cable system inspection, cable maintenance, fittings, pulleys and wear patterns, Construction and servicing of Bowden cable and Teleflex controls
5. Specific Objectives	9.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Inspection of Bowden cable assembly, identification of wear patterns on pulleys, verification of fitting torque standards, safety method demonstration for cable tensioning, instructor supervision of trainee cable servicing procedure
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of control cables in an aircraft system?

(b) List three common types of cable fittings used in aviation maintenance.

(c) Why is regular inspection of cable wear patterns critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Cable system inspection procedures • Identification of bolts and nuts standards • Safety methods for cable handling • Pulley wear pattern analysis • Fitting torque specifications	L & Ds	PPT, IB	15 min
(b)	• Construction of Bowden cable assemblies • Servicing procedures for Teleflex controls • Identification of corrosion and damage • Replacement criteria for worn cables • Proper lubrication and tensioning techniques	L & Ds	PPT, PDF	20 min
(c)	• Documentation of maintenance actions • Reporting wear patterns and defects • Final verification of cable tension • Storage and handling of spare cables • Review of safety protocols during servicing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between internal and external cable wear patterns.

(b) Describe the correct procedure for identifying and replacing a damaged cable fitting.

(c) How do you verify that a serviced Bowden cable meets tension standards?

12. Home Assignment and Follow Up	Prepare a brief report on the key differences between Bowden cable and Teleflex control systems, referencing standards and maintenance requirements for the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various locking devices, Fasteners and fork joints	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min